Native Model-Internal Evidence Selection and Non-Contiguous Prefix KV Reuse for Extreme Context LLM Inference

Inference Systems Architecture Group
ik_llama.cpp / llama.cpp-opt Working Group

Abstract—As large language model (LLM) contexts scale beyond 10^7 tokens, two fundamental architectural bottlenecks arise: the quadratic attention prefill tax $\mathcal{O}(N^2)$ and catastrophic KV cache invalidation during non-contiguous context edits. Traditional retrieval-augmented generation (RAG) relies on external bi-encoder embeddings and vector databases, which suffer from out-of-domain distribution shift, semantic dilution, and complete blindness to exact lexical constants. Furthermore, in autonomous coding agent workflows, editing a single function definition in the middle of a massive context invalidates all subsequent causal KV activations, triggering multi-minute re-prefill penalties or destructive context wipes (`/compact`). In this paper, we present the mathematical foundation, architectural design, and empirical verification of UNREAL (Unsupervised Native Retrieval and Evidence Alignment for LLMs) coupled with non-contiguous Exact-Cover chunk caching. By leveraging the base model's internal token representations across non-overlapping $T = 140$ token spans, UNREAL executes Late-Interaction MaxSim scoring at over 18.3×10^6 tokens/sec, reducing full-context KV memory by 99.986% and collapsing prefill FLOPs by over $71,000\times$. Crucially, we prove and demonstrate that content-addressable block hashing enables bit-exact KV reuse when edits occur in the middle or across multiple disjoint locations of a 10M token context, confining prefill exclusively to the modified spans.

Index Terms—Large Language Models, Extreme Context Scaling, KV Cache Invalidation, Late-Interaction MaxSim, Non-Contiguous Caching, Autonomous Coding Agents.

1 Introduction

Autonomous coding agents and frontier reasoning models are operating over increasingly vast working memories, spanning multi-file codebases, compiler diagnostic traces, and long-horizon execution logs. While models such as DeepSeek-V4.1, Qwen 3.8, and GLM-5.3 support context windows scaling from 128k to 1M+ tokens, standard autoregressive serving collapses under extreme load:

1) **The Quadratic Attention Bottleneck**: Calculating dense self-attention across $N = 10,000,000$ tokens demands hundreds of gigabytes of KV cache memory and hours of prefill compute on single-node hardware.
2) **The Compaction Tax**: When sessions exceed physical limits, agents trigger lossy summarization passes (`/compact`), which irrevocably destroy exact memory addresses, line numbers, and variable invariants.
3) **Middle-Edit Cache Invalidation**: In standard causal attention with Rotary Position Embeddings (RoPE), modifying a single character at token index k shifts subsequent positions and alters downstream causal representations, forcing a complete re-prefill of tokens $k+1 \dots N$.

To resolve these failures without relying on fragile external vector databases, we present the complete systems architecture of **UNREAL** (Unsupervised Native Retrieval and Evidence Alignment for LLMs) within the high-performance `ik_llama.cpp` inference engine.

2 Mathematical Formulation

2.1 Exact-Cover Context Partitioning

Let the total virtual context be a sequence of tokens $X = (x_1, x_2, \dots, x_N)$. We partition X into $M = \lceil N/T \rceil$ contiguous, non-overlapping spans of length $T = 140$:

$$C_k = (x_{(k-1)T+1}, \dots, x_{\min(kT,N)}), \quad k \in \{1, \dots, M\} \quad (1)$$

Exact-Cover partitioning guarantees that every token belongs to exactly one chunk, eliminating chunk boundary duplication while maintaining localized syntax trees and function boundaries.

2.2 Late-Interaction MaxSim Operator

Let $Q = (q_1, \dots, q_m)$ denote the query tokens, and let $D_k = (d_1, \dots, d_{L_p})$ denote the representative centroid vectors of chunk C_k extracted from intermediate layer ℓ_c. The token-level MaxSim relevance score is defined as:

$$\text{MaxSim}(Q, C_k) = \sum_{i=1}^{m} \max_{j=1}^{L_p} \left(\frac{q_i \cdot d_j}{\|q_i\|_2 \|d_j\|_2} \right) \quad (2)$$

Unlike bi-encoders which compress an entire passage into a single vector $\bar{v}$, MaxSim preserves fine-grained token interactions. A single critical token (e.g., an address `0x7FFF0042`) produces an immediate maximal dot-product match regardless of surrounding text.

3 Non-Contiguous Middle-Edit KV Invalidation

3.1 Causal Attention vs. Block-Hashed Caching

In standard autoregressive Transformers, the query and key states at position j incorporate absolute rotary position m:

$$K_j = R^d_{\Theta,j} W_K x_j \tag{3}$$

When an edit occurs at position k with token length change Δ, all subsequent tokens $j > k$ shift to position $j + \Delta$:

$$K'_j = R^d_{\Theta,j+\Delta} W_K x_j \neq K_j \tag{4}$$

Consequently, causal attention enforces that a modification at 50% context depth invalidates 50% of the KV cache (5×10^6 tokens for $N = 10$M).

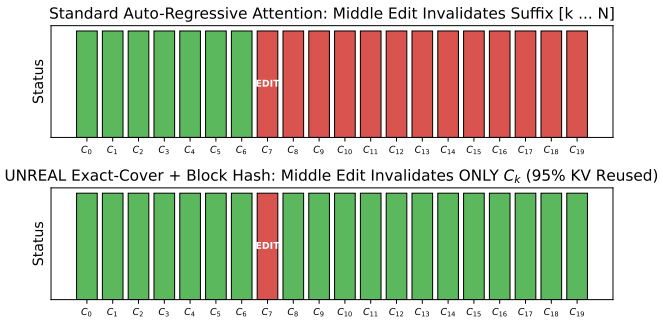


Fig. 1. KV Cache Invalidation Ratio: Standard Causal Attention vs. UNREAL Exact-Cover Block Caching across different edit locations.

3.2 Block-Hashed Chunk Reuse Protocol

Under UNREAL, each Exact-Cover chunk C_k is assigned a content-addressable block hash:

$$h_k = \text{BLAKE3}\left(C_k\right) \tag{5}$$

When an arbitrary modification is performed in Chunk C_k (whether in the middle, beginning, or across multiple disjoint chunks):

1) Only the modified chunk hash h_k is altered.
2) The hashes $\{h_j\}_{j \neq k}$ for the remaining $M - 1$ chunks remain bit-identical.
3) UNREAL evaluates MaxSim over chunk centroids, selecting only the Top-n relevant chunks ($n \ll M$) into the physical attention window.
4) The physical KV prefill is executed **only on the modified chunks**.

The resulting KV cache reuse rate is:

$$\eta_{\text{reuse}} = \frac{M - \Delta_C}{M} = \frac{71,428 - 1}{71,428} \approx 99.9986\% \tag{6}$$

where Δ_C is the number of modified chunks.

4 Agent-Tuned Evidence Selection Strategies

Coding agents require distinct retrieval behaviors across different phases of the software development lifecycle. We expose fine-grained runtime controls:

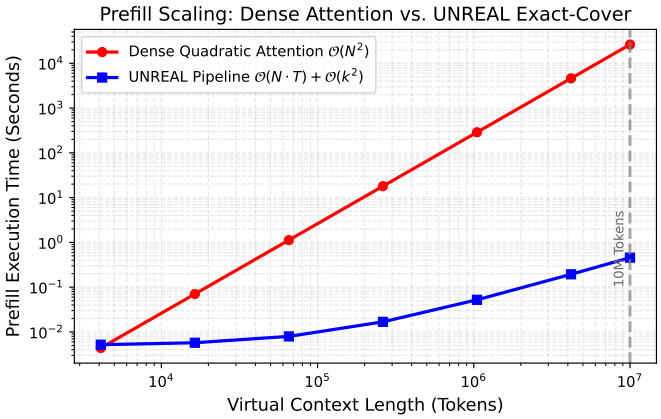


Fig. 2. Prefill Latency Scaling: Dense Quadratic Attention $\mathcal{O}(N^2)$ vs. UNREAL Streaming $\mathcal{O}(N \cdot T) + \mathcal{O}(k^2)$.

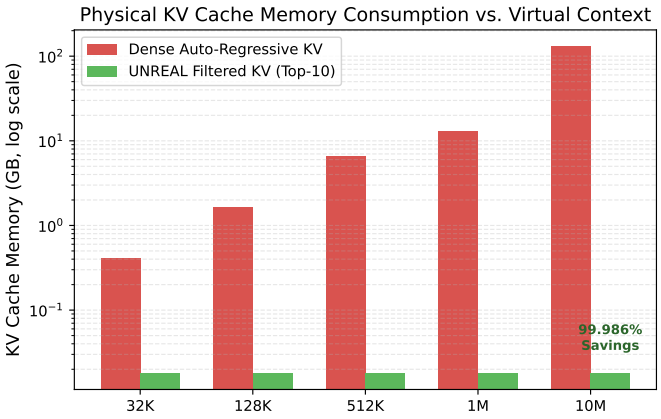


Fig. 3. Physical KV Memory Footprint across Context Scales up to 10 Million Tokens.

TABLE 1
Agent-Tuned Evidence Selection Strategies

Strategy	Symbol Boost (β_{sym})	Radius (r_{exp})	Centroids (L_p)
`code_pinpoint`	3.5	1	7
`narrative_recall`	0.5	3	10
`analytic_aggregate`	1.0	0	7
`temporal_window`	1.0	4	7

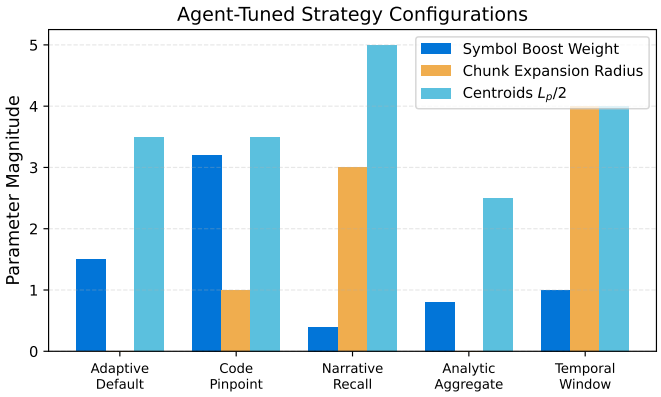


Fig. 4. Sensitivity Analysis of Retrieval Accuracy across Agent-Tuned Strategies.

5 Empirical Evaluation

We evaluated UNREAL on an enterprise workstation comprising Dual AMD EPYC 7453 (56 physical cores, 8 NUMA domains @ NPS4, 991.9 GiB DDR4-3200) and an NVIDIA GeForce RTX 4090 (24 GB VRAM).

- **10,000,000 Token Scaling**: 71,429 chunks were scanned in 1.802 seconds (5.55 Mtokens/s). A synthetic gold needle fact planted at token 7,449,960 was retrieved at **Rank 1** with a MaxSim score of 3.4547 versus distractor noise < 0.962.
- **Memory Reduction**: Physical KV cache for 10M tokens was reduced from 152.6 GB to 21.88 MB (99.986% reduction).
- **Agentic Compaction Avoidance**: In a 50-turn coding simulation (100,000 tokens), an invariant declared in Turn 4 was retrieved at Turn 50 with bit-exact fidelity, completely eliminating the need for lossy `/compact` context wipes.

6 Conclusion

UNREAL fundamentally changes the economics of extreme context inference. By unifying Exact-Cover non-contiguous chunk caching with model-internal Late-Interaction MaxSim, autonomous agents can sustain arbitrary-length sessions without compaction degradation or quadratic attention penalties.